

Creation Date 22-Sep-2009

Revision Date 26-Jan-2024

Revision Number 5

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                                 |
|---------------------------|---------------------------------|
| Product Description:      | <b>Cobalt sputtering target</b> |
| Cat No. :                 | <b>40837</b>                    |
| Index No                  | 027-001-00-9                    |
| CAS No                    | 7440-48-4                       |
| EC No                     | 231-158-0                       |
| Molecular Formula         | Co                              |
| REACH registration number | -                               |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

# SAFETY DATA SHEET

Cobalt sputtering target

Revision Date 26-Jan-2024

## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Based on available data, the classification criteria are not met

### Health hazards

Acute oral toxicity  
Respiratory Sensitization  
Skin Sensitization  
Germ Cell Mutagenicity  
Carcinogenicity  
Reproductive Toxicity

Category 4 (H302)  
Category 1 Sub-category 1B (H334)  
Category 1 (H317)  
Category 2 (H341)  
Category 1B (H350i)  
Category 1B (H360F)

### Environmental hazards

Chronic aquatic toxicity

Category 3 (H412)

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

**Danger**

### **Hazard Statements**

H302 - Harmful if swallowed  
H317 - May cause an allergic skin reaction  
H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled  
H341 - Suspected of causing genetic defects  
H350i - May cause cancer by inhalation  
H360F - May damage fertility  
H412 - Harmful to aquatic life with long lasting effects

### **Precautionary Statements**

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention  
P284 - Wear respiratory protection  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### **Additional EU labelling**

Restricted to professional users

## 2.3. Other hazards

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment

# SAFETY DATA SHEET

Cobalt sputtering target

Revision Date 26-Jan-2024

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component       | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                        |
|-----------------|-----------|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cobalt, massive | 7440-48-4 | EEC No. 231-158-0 | <=100    | Acute Tox. 4 (H302)<br>Resp. Sens. 1B (H334)<br>Skin Sens. 1 (H317)<br>Muta.2 (H341)<br>Repr. 1B (H360F)<br>Carc. 1B (H350i)<br>Aquatic Chronic 3 (H412) |

REACH registration number

-

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | If symptoms persist, call a physician.                                                                                                           |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| Ingestion                          | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                            |
| Inhalation                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable. . May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Notes to Physician | Treat symptomatically. |
|--------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

Suitable Extinguishing Media  
approved class D extinguishers.

# SAFETY DATA SHEET

Cobalt sputtering target

Revision Date 26-Jan-2024

## Extinguishing media which must not be used for safety reasons

Water may be ineffective.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

## Hazardous Combustion Products

Cobalt oxides.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. No special precautions required.

### 6.2. Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system.

### 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Pick up and transfer to properly labelled containers.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep in a dry place. Keep away from acids.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 6.1D

**Switzerland - Storage of hazardous substances**

Storage class - SC 6.1  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

# SAFETY DATA SHEET

Cobalt sputtering target

Revision Date 26-Jan-2024

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component       | European Union | The United Kingdom                                                                   | France | Belgium                               | Spain                                             |
|-----------------|----------------|--------------------------------------------------------------------------------------|--------|---------------------------------------|---------------------------------------------------|
| Cobalt, massive |                | STEL: 0.3 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr<br>Resp. Sens. |        | TWA: 0.02 mg/m <sup>3</sup> 8<br>uren | TWA / VLA-ED: 0.02<br>mg/m <sup>3</sup> (8 horas) |

| Component       | Italy | Germany | Portugal                               | The Netherlands                       | Finland                                   |
|-----------------|-------|---------|----------------------------------------|---------------------------------------|-------------------------------------------|
| Cobalt, massive |       | Haut    | TWA: 0.02 mg/m <sup>3</sup> 8<br>horas | TWA: 0.02 mg/m <sup>3</sup> 8<br>uren | TWA: 0.02 mg/m <sup>3</sup> 8<br>tunteina |

| Component       | Austria                                                                                                                                                                  | Denmark                                                                               | Switzerland                                           | Poland                                     | Norway                                                                                                          |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Cobalt, massive | TRK-KZGW: 2 mg/m <sup>3</sup><br>15 Minuten<br>TRK-KZGW: 0.4 mg/m <sup>3</sup><br>15 Minuten<br>Haut<br>TRK-TMW: 0.5 mg/m <sup>3</sup><br>TRK-TMW: 0.1 mg/m <sup>3</sup> | TWA: 0.01 mg/m <sup>3</sup> 8<br>timer<br>STEL: 0.02 mg/m <sup>3</sup> 15<br>minutter | Haut/Peau<br>TWA: 0.05 mg/m <sup>3</sup> 8<br>Stunden | TWA: 0.02 mg/m <sup>3</sup> 8<br>godzinach | TWA: 0.02 mg/m <sup>3</sup> 8<br>timer<br>STEL: 0.06 mg/m <sup>3</sup> 15<br>minutter. value<br>calculated fume |

| Component       | Bulgaria                   | Croatia                                     | Ireland                                                                 | Cyprus | Czech Republic                                                                                                |
|-----------------|----------------------------|---------------------------------------------|-------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------|
| Cobalt, massive | TWA: 0.1 mg/m <sup>3</sup> | TWA-GVI: 0.1 mg/m <sup>3</sup> 8<br>satima. | TWA: 0.02 mg/m <sup>3</sup> 8 hr.<br>STEL: 0.3 mg/m <sup>3</sup> 15 min |        | TWA: 0.05 mg/m <sup>3</sup> 8<br>hodinách. inhalable<br>fraction of aerosol<br>Ceiling: 0.1 mg/m <sup>3</sup> |

| Component       | Estonia                                    | Gibraltar | Greece                     | Hungary                                     | Iceland                                                                                                              |
|-----------------|--------------------------------------------|-----------|----------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Cobalt, massive | TWA: 0.05 mg/m <sup>3</sup> 8<br>tundides. |           | TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.02 mg/m <sup>3</sup> 8<br>óraban. AK | TWA: 0.02 mg/m <sup>3</sup> 8<br>klukkustundum. dust<br>and fume<br>Ceiling: 0.04 mg/m <sup>3</sup><br>dust and fume |

| Component       | Latvia                     | Lithuania                        | Luxembourg | Malta | Romania                                                                       |
|-----------------|----------------------------|----------------------------------|------------|-------|-------------------------------------------------------------------------------|
| Cobalt, massive | TWA: 0.5 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> IPRD |            |       | TWA: 0.05 mg/m <sup>3</sup> 8 ore<br>STEL: 0.1 mg/m <sup>3</sup> 15<br>minute |

| Component       | Russia                                                                           | Slovak Republic             | Slovenia | Sweden                                              | Turkey |
|-----------------|----------------------------------------------------------------------------------|-----------------------------|----------|-----------------------------------------------------|--------|
| Cobalt, massive | TWA: 0.01 mg/m <sup>3</sup> 1108<br>Skin notation<br>MAC: 0.05 mg/m <sup>3</sup> | TWA: 0.05 mg/m <sup>3</sup> |          | TLV: 0.02 mg/m <sup>3</sup> 8<br>timmar. NGV<br>Hud |        |

#### Biological limit values

List source(s):

| Component       | European Union | United Kingdom | France                                                                                                                           | Spain                                                                               | Germany |
|-----------------|----------------|----------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------|
| Cobalt, massive |                |                | Cobalt: 0.001 mg/L<br>blood end of shift at end<br>of workweek<br>Cobalt: 0.015 mg/L<br>urine end of shift at end<br>of workweek | Cobalt: 15 µg/L urine<br>end of workweek<br>Cobalt: 1 µg/L blood<br>end of workweek |         |

| Component | Italy | Finland | Denmark | Bulgaria | Romania |
|-----------|-------|---------|---------|----------|---------|
|-----------|-------|---------|---------|----------|---------|

# SAFETY DATA SHEET

Cobalt sputtering target

Revision Date 26-Jan-2024

|                 |  |                                                                                                 |  |  |                                                                                 |
|-----------------|--|-------------------------------------------------------------------------------------------------|--|--|---------------------------------------------------------------------------------|
| Cobalt, massive |  | Cobalt: 130 nmol/L urine after the work phase or shift after a working week or exposure period. |  |  | Cobalt: 15 µg/L urine end of work week<br>Cobalt: 1 µg/L blood end of work week |
|-----------------|--|-------------------------------------------------------------------------------------------------|--|--|---------------------------------------------------------------------------------|

| Component       | Gibraltar | Latvia | Slovak Republic                    | Luxembourg | Turkey |
|-----------------|-----------|--------|------------------------------------|------------|--------|
| Cobalt, massive |           |        | Cobalt: 30 µg/L urine not critical |            |        |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

MDHS30/2 Cobalt and cobalt compounds in air Laboratory method using flame atomic absorption spectrometry

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                             | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|---------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Cobalt, massive<br>7440-48-4 ( ≤100 ) |                                  |                                     | DNEL = 40µg/m³                     |                                       |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                             | Fresh water     | Fresh water sediment         | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)       |
|---------------------------------------|-----------------|------------------------------|--------------------|------------------------------------|--------------------------|
| Cobalt, massive<br>7440-48-4 ( ≤100 ) | PNEC = 0.62µg/L | PNEC = 53.8mg/kg sediment dw |                    | PNEC = 0.37mg/L                    | PNEC = 10.9mg/kg soil dw |

| Component                             | Marine water    | Marine water sediment        | Marine water Intermittent | Food chain | Air |
|---------------------------------------|-----------------|------------------------------|---------------------------|------------|-----|
| Cobalt, massive<br>7440-48-4 ( ≤100 ) | PNEC = 2.36µg/L | PNEC = 69.8mg/kg sediment dw |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. None under normal use conditions.

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

No special protective equipment required

| Glove material    | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Disposable gloves | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

#### Respiratory Protection

No special protective equipment required.

# SAFETY DATA SHEET

Cobalt sputtering target

Revision Date 26-Jan-2024

|                                 |                                                                                                                              |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Large scale/emergency use       | In case of insufficient ventilation, wear suitable respiratory equipment                                                     |
| Small scale/Laboratory use      | No personal respiratory protective equipment normally required<br>When RPE is used a face piece Fit Test should be conducted |
| Environmental exposure controls | Prevent product from entering drains.                                                                                        |

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Physical State                          | Solid                    |                                   |
| Appearance                              | Grey                     |                                   |
| Odor                                    | Odorless                 |                                   |
| Odor Threshold                          | No data available        |                                   |
| Melting Point/Range                     | 1495 °C / 2723 °F        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | 2870 °C / 5198 °F        |                                   |
| Flammability (liquid)                   | Not applicable           | Solid                             |
| Flammability (solid,gas)                | No information available |                                   |
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | Not applicable           | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | No information available | 4444                              |
| Viscosity                               | Not applicable           | Solid                             |
| Water Solubility                        | Insoluble                |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| Cobalt, massive                         | 5                        |                                   |
| Vapor Pressure                          | No information available |                                   |
| Density / Specific Gravity              | 8.92                     |                                   |
| Bulk Density                            | No data available        |                                   |
| Vapor Density                           | Not applicable           | Solid                             |
| Particle characteristics                | No data available        |                                   |

### 9.2. Other information

|                   |                        |
|-------------------|------------------------|
| Molecular Formula | Co                     |
| Molecular Weight  | 58.9332                |
| Evaporation Rate  | Not applicable - Solid |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                               |
|--------------------------|-------------------------------|
| Hazardous Polymerization | No information available.     |
| Hazardous Reactions      | None under normal processing. |

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

# SAFETY DATA SHEET

Cobalt sputtering target

Revision Date 26-Jan-2024

## 10.5. Incompatible materials

None known.

## 10.6. Hazardous decomposition products

Cobalt oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral Category 4  
Dermal No data available  
Inhalation No data available

| Component       | LD50 Oral                           | LD50 Dermal | LC50 Inhalation              |
|-----------------|-------------------------------------|-------------|------------------------------|
| Cobalt, massive | 6171 mg/kg (Rat)<br>550 mg/kg (Rat) | -           | LC50 < 0.05 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

Respiratory Sub Category 1B  
Skin Category 1  
No information available

(e) germ cell mutagenicity; Category 2

(f) carcinogenicity; No data available

This product contains one or more substances which are classified by IARC as carcinogenic to humans (Group I), probably carcinogenic to humans (Group 2A) or possibly carcinogenic to humans (Group 2B) The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component       | EU           | UK | Germany | IARC     |
|-----------------|--------------|----|---------|----------|
| Cobalt, massive | Carc Cat. 1B |    | Cat. 2  | Group 2A |

##### (g) reproductive toxicity; Reproductive Effects

Category 1B  
Product is or contains a chemical which is a known or suspected reproductive hazard.

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; Not applicable  
Solid

Symptoms / effects, both acute and Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling



# SAFETY DATA SHEET

Cobalt sputtering target

Revision Date 26-Jan-2024

delayed of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system. Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component       | Freshwater Fish                                  | Water Flea | Freshwater Algae |
|-----------------|--------------------------------------------------|------------|------------------|
| Cobalt, massive | LC50: > 100 mg/L, 96h static (Brachydanio rerio) |            |                  |

### 12.2. Persistence and degradability

#### Persistence Degradability Degradation in sewage treatment plant

Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
May persist.  
Not relevant for inorganic substances.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

Product has a high potential to bioconcentrate

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| Cobalt, massive | 5       | No data available             |

### 12.4. Mobility in soil

Spillage unlikely to penetrate soil Is not likely mobile in the environment due its low water solubility. Is not likely mobile in the environment due its low water solubility and propensity to bind to soil particles

### 12.5. Results of PBT and vPvB assessment

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

### 12.6. Endocrine disrupting properties

#### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

# SAFETY DATA SHEET

Cobalt sputtering target

Revision Date 26-Jan-2024

|                                       |                                                                                                                                                                                                                                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>European Waste Catalogue (EWC)</b> | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>              | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Do not let this chemical enter the environment.                                                                                                       |
| <b>Switzerland - Waste Ordinance</b>  | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO** Not regulated

14.1. UN number  
14.2. UN proper shipping name  
14.3. Transport hazard class(es)  
14.4. Packing group

**ADR** Not regulated

14.1. UN number  
14.2. UN proper shipping name  
14.3. Transport hazard class(es)  
14.4. Packing group

**IATA** Not regulated

14.1. UN number  
14.2. UN proper shipping name  
14.3. Transport hazard class(es)  
14.4. Packing group

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component       | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Cobalt, massive | 7440-48-4 | 231-158-0 | -      | -   | X     | X    | KE-06060 | X    | -    |

| Component       | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Cobalt, massive | 7440-48-4 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

# SAFETY DATA SHEET

Cobalt sputtering target

Revision Date 26-Jan-2024

## Authorisation/Restrictions according to EU REACH

| Component       | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                  | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|-----------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Cobalt, massive | 7440-48-4 | -                                                                   | Use restricted. See item 30.<br>(see link for restriction details)<br>Use restricted. See item 28.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |

### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component       | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Cobalt, massive | 7440-48-4 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component       | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                                                                                                                   |
|-----------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Cobalt, massive | WGK 3                                 | Class II : 0.5 mg/m <sup>3</sup> (Massenkonzentration)<br>Krebserzeugende Stoffe - Class I : 0.05 mg/m <sup>3</sup> (Massenkonzentration) |

| Component       | France - INRS (Tables of occupational diseases)                              |
|-----------------|------------------------------------------------------------------------------|
| Cobalt, massive | Tableaux des maladies professionnelles (TMP) - RG 65,RG 70,RG 70bis,RG 70ter |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

# SAFETY DATA SHEET

Cobalt sputtering target

Revision Date 26-Jan-2024

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed  
H317 - May cause an allergic skin reaction  
H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled  
H341 - Suspected of causing genetic defects  
H350 - May cause cancer  
H350i - May cause cancer by inhalation  
H360F - May damage fertility  
H412 - Harmful to aquatic life with long lasting effects

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

### Prepared By

Health, Safety and Environmental Department

### Creation Date

22-Sep-2009

### Revision Date

26-Jan-2024

### Revision Summary

New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

# **SAFETY DATA SHEET**

**Cobalt sputtering target**

**Revision Date** 26-Jan-2024

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date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**